A Simple Method for Power System Stability Analysis with Multiple Time Delays

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***Abstract*—There exist significant time delays in the data of phasor measurement unit (PMU) and wide-area measurement system (WAMS). To properly evaluate the impact of time delay is very important for power system online stability assessment and controller design. In this paper, the maximum delayed time that power system can sustain without losing its stability is denoted as its delay margin. From the viewpoint of Lyapunov stability, when a dynamic system is critical stable at a given equilibrium point, there must exist critical eigenvalue on the imaginary axis. This paper presents a simple method to determine such critical eigenvalue by tracing eigenvalue loci of a transformation matrix. Then the critical eigenvalue is used to determine the delay margin of power system. Comparing with previous approaches, such as Lyapunov function/functional method, LMI method, root clustering paradigm method, the presented method can locate the exact delay margin with no conservativeness and with less computation burden. Moreover, it can be easily applied to the stability analysis of bulk power system with multiple time delays. Finally, single machine and infinite bus system (SMIB) and WSCC 3-generator-9-bus system are employed to validate its effectiveness.**

***Index Terms*—Stability analysis, time delay, power system**

I. INTRODUCTION

IN the physical world, system trend is not only dependent on the current operation point but also subject to its previous conditions. This phenomenon is denoted as the time delay [1]. Although time delay exists widely in the power system measurement and control loops, it was usually ignored in the past stability study and controller design. The reason was that the traditional power system controllers were generally designed based on the local information. And, time delay of the local measurement was small enough to be ignored with little error. However, time delay is significant (from tens milliseconds to hundreds milliseconds) in the wide-area environment, it can not be simply ignored. So it is important to properly evaluate the impact of time delay on power system stability assessment and controller design.

[2-5] discussed the influence of time delay on the control of PSS, TCSC, SVC and UPFC. [6, 7] investigated the cause of time delay in power system and aimed to find more effective compensation methods. [8, 9] applied delayed feedback control to eliminate periodic and chaotic oscillation in power system operation. [10, 11] analyzed the impact of time delay on power system small signal stability and small signal stability region.

In power system wide-area stability assessment and controller design, an essential issue is to know the maximum delayed time that power system can sustain without losing its stability. In this paper, this value is denoted as the delay margin of power system. Since the characteristic equation of a time-delayed system is transcendental, it is hard to be solved directly. [12] gave a method to determine the delay margin based on Schur-Cohu-Fujiwara theory. [13] used an optimal procedure to locate the critical eigenvalue of a time-delayed system and then determined its delay margin. [14, 15] gave another method based on Rekasius substitution and root clustering paradigm theory. However, such methods can not be used to the delay margin analysis of bulk power system for their complexity and heavy computation. [16-20] gave some approaches to approximate the delay margin by Lyapunov stability theory and LMI method. However, there exists significant conservativeness in such methods.

In this paper, a simple method to determine the delay margin of power system is presented. It traces the eigenvalue loci of a transformation matrix so as to locate all the critical eigenvalues of the time-delayed system, which are used to calculate the delay margin. Comparing with the previous methods of [12-20], it can determine the exact delay margin without any conservativeness. And, its computation burden is relatively lower. So, it can be applied to the delay margin analysis of bulk power system.

Rest of this paper is organized as following: Section 2 gives a brief review to the delay margin definition and power system model. Section 3 derives the algorithm for delay margin determination. Case studies and conclusions are given in section 4 and section 5.

II. DELAY MARGIN DEFINITION

2.1. Power System Stability without Time Delay

Power system dynamics are usually described by the following differential-algebraic equation (DAE) when there is no time delay.

$$\begin{aligned} \dot{\boldsymbol{x}} &= \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{p}) \\ \boldsymbol{0} &= \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{p}) \end{aligned} \tag{1}$$

Where, $\boldsymbol{x} \in R^n$, $\boldsymbol{y} \in R^m$, $\boldsymbol{p} \in R^p$ are the vectors of state variables, algebraic variables and parameter variables. For a given $\boldsymbol{p}$, equilibrium point of Eq.(1) is the solution of the following equation.

$$\begin{aligned} \boldsymbol{0} &= \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y})\big|_{\boldsymbol{p}} \\ \boldsymbol{0} &= \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{y})\big|_{\boldsymbol{p}} \end{aligned} \tag{2}$$

Linearizing Eq.(1) at an equilibrium point $(\boldsymbol{x}_0, \boldsymbol{y}_0)$, we can get the incremental differential-algebraic equation as below.

This work is supported by the NSFC (50707019, 50595413), Special Fund of the National Fundamental Research (2004CB217904), FANEDD (200439), Program for New Century Excellent Talents in University, Fok Ying Tung Education Foundation (104019) and Tianjin Municipal (06TXTJJC13700) of China

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$$\begin{aligned} \Delta\dot{\boldsymbol{x}} &= \boldsymbol{A}(\boldsymbol{p})\cdot\Delta\boldsymbol{x} + \boldsymbol{B}(\boldsymbol{p})\cdot\Delta\boldsymbol{y} \\ \boldsymbol{0} &= \boldsymbol{C}(\boldsymbol{p})\cdot\Delta\boldsymbol{x} + \boldsymbol{D}(\boldsymbol{p})\cdot\Delta\boldsymbol{y} \end{aligned} \tag{3}$$

Where, $\boldsymbol{A}(\boldsymbol{p}) = \partial\boldsymbol{f}/\partial\boldsymbol{x}\big|_{\boldsymbol{p}}$, $\boldsymbol{B}(\boldsymbol{p}) = \partial\boldsymbol{f}/\partial\boldsymbol{y}\big|_{\boldsymbol{p}}$, $\boldsymbol{C}(\boldsymbol{p}) = \partial\boldsymbol{g}/\partial\boldsymbol{x}\big|_{\boldsymbol{p}}$, $\boldsymbol{D}(\boldsymbol{p}) = \partial\boldsymbol{g}/\partial\boldsymbol{y}\big|_{\boldsymbol{p}}$. If matrix $\boldsymbol{D}$ is nonsingular, Eq.(3) can be reduced into the following ordinary differential equation.

$$\Delta\dot{\boldsymbol{x}} = \tilde{\boldsymbol{A}}(\boldsymbol{p})\cdot\Delta\boldsymbol{x} \tag{4}$$

Where, $\tilde{\boldsymbol{A}}(\boldsymbol{p}) = \boldsymbol{A}(\boldsymbol{p}) - \boldsymbol{B}(\boldsymbol{p})\cdot\boldsymbol{D}^{-1}(\boldsymbol{p})\cdot\boldsymbol{C}(\boldsymbol{p})$ (5)

Further we denote $\boldsymbol{\Lambda} = (\lambda_1, \lambda_2, ..., \lambda_n)$ as the eigenvalue spectrum of matrix $\tilde{\boldsymbol{A}}(\boldsymbol{p})$ and use C^-, C^+, C^0 for the left half, right half and imaginary axis of the complex plane respectively. Stability of the equilibrium point $(\boldsymbol{x}_0, \boldsymbol{y}_0)$ then can be judged by the eigenvalues of matrix $\tilde{\boldsymbol{A}}(\boldsymbol{p})$: If all eigenvalues stay in C^-, $(\boldsymbol{x}_0, \boldsymbol{y}_0)$ is stable. If we can find some eigenvalue in C^+, $(\boldsymbol{x}_0, \boldsymbol{y}_0)$ is unstable. Once there are several eigenvalues on C^0 and all rest stay in C^-, $(\boldsymbol{x}_0, \boldsymbol{y}_0)$ is critical stable. In power system daily operation, it is essential to keep the operating point far away from the critical stable and unstable condition, .i.e. to keep all the eigenvalues of matrix $\tilde{\boldsymbol{A}}(\boldsymbol{p})$ in the left half plane.

2.2. Power System Stability with Time Delays

If there exist time delays, power system dynamics have to be described by the following time-delayed differential-algebraic equation (TDAE).

$$\begin{cases} \dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{x}_{\tau 1}, \boldsymbol{y}_{\tau 1}, \boldsymbol{x}_{\tau 2}, \boldsymbol{y}_{\tau 2}, ..., \boldsymbol{x}_{\tau m}, \boldsymbol{y}_{\tau m}, \boldsymbol{p}) \\ \boldsymbol{0} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{p}) \\ \boldsymbol{0} = \boldsymbol{g}(\boldsymbol{x}_{\tau 1}, \boldsymbol{y}_{\tau 1}, \boldsymbol{p}) \\ \boldsymbol{0} = \boldsymbol{g}(\boldsymbol{x}_{\tau 2}, \boldsymbol{y}_{\tau 2}, \boldsymbol{p}) \\ \ldots\ \ldots \\ \boldsymbol{0} = \boldsymbol{g}(\boldsymbol{x}_{\tau m}, \boldsymbol{y}_{\tau m}, \boldsymbol{p}) \end{cases} \tag{6}$$

Where, $\boldsymbol{x} \in R^n$, $\boldsymbol{y} \in R^m$, $\boldsymbol{p} \in R^p$ are the same as Eq.(1). $(\boldsymbol{x}_{\tau i}, \boldsymbol{y}_{\tau i}) := [\boldsymbol{x}(t-\tau_i), \boldsymbol{y}(t-\tau_i)], i = 1,2,...,m$ are the vectors of time-delayed state variables and algebraic variables. $\tau_i > 0$, i=1,2,3…,m. are the time-delayed constants. They are considered independently in this paper for simplification. We denote $\boldsymbol{T} = (\tau_1, \tau_2, ..., \tau_m)$ as the vector of time delays. Suppose that $(\boldsymbol{x}_0, \boldsymbol{y}_0)$ is an equilibrium point, where we can linearize Eq.(6) and get the following incremental equation.

$$\begin{cases} \Delta\dot{\boldsymbol{x}} = \boldsymbol{A}_0\Delta\boldsymbol{x} + \boldsymbol{B}_0\Delta\boldsymbol{y} + \sum_{i=1}^{m}(\boldsymbol{A}_{\tau i}\Delta\boldsymbol{x}_{\tau i} + \boldsymbol{B}_{\tau i}\Delta\boldsymbol{y}_{\tau i}) \\ \boldsymbol{0} = \boldsymbol{C}_0\Delta\boldsymbol{x} + \boldsymbol{D}_0\Delta\boldsymbol{y} \\ \boldsymbol{0} = \boldsymbol{C}_{\tau 1}\Delta\boldsymbol{x}_{\tau 1} + \boldsymbol{D}_{\tau 1}\Delta\boldsymbol{y}_{\tau 1} \\ \boldsymbol{0} = \boldsymbol{C}_{\tau 2}\Delta\boldsymbol{x}_{\tau 2} + \boldsymbol{D}_{\tau 2}\Delta\boldsymbol{y}_{\tau 2} \\ \ldots\ \ldots \\ \boldsymbol{0} = \boldsymbol{C}_{\tau m}\Delta\boldsymbol{x}_{\tau m} + \boldsymbol{D}_{\tau m}\Delta\boldsymbol{y}_{\tau m} \end{cases} \tag{7}$$

Where, $\boldsymbol{A}_0 = \partial\boldsymbol{f}/\partial\boldsymbol{x}\big|_{\boldsymbol{p}}, \boldsymbol{B}_0 = \partial\boldsymbol{f}/\partial\boldsymbol{y}\big|_{\boldsymbol{p}}, \boldsymbol{C}_0 = \partial\boldsymbol{g}/\partial\boldsymbol{x}\big|_{\boldsymbol{p}}, \boldsymbol{D}_0 = \partial\boldsymbol{g}/\partial\boldsymbol{y}\big|_{\boldsymbol{p}}$, $\boldsymbol{A}_{\tau i} = \partial\boldsymbol{f}/\partial\boldsymbol{x}_{\tau i}\big|_{\boldsymbol{p}}, \boldsymbol{B}_{\tau i} = \partial\boldsymbol{f}/\partial\boldsymbol{y}_{\tau i}\big|_{\boldsymbol{p}}, \boldsymbol{C}_{\tau i} = \partial\boldsymbol{g}/\partial\boldsymbol{x}_{\tau i}\big|_{\boldsymbol{p}}, \boldsymbol{D}_{\tau i} = \partial\boldsymbol{g}/\partial\boldsymbol{y}_{\tau i}\big|_{\boldsymbol{p}}$.

If matrices $\boldsymbol{D}_0, \boldsymbol{D}_{\tau i}$ are nonsingular, Eq.(7) can be reduced into the following ordinary differential equation:

$$\Delta\dot{\boldsymbol{x}} = \tilde{\boldsymbol{A}}_0\Delta\boldsymbol{x} + \sum_{i=1}^{m}\tilde{\boldsymbol{A}}_{\tau i}\Delta\boldsymbol{x}_{\tau i} \tag{8}$$

With: $\tilde{\boldsymbol{A}}_0 = \boldsymbol{A}_0 - \boldsymbol{B}_0\cdot\boldsymbol{D}_0^{-1}\cdot\boldsymbol{C}_0$ (9a)

$$\tilde{\boldsymbol{A}}_{\tau i} = \boldsymbol{A}_{\tau i} - \boldsymbol{B}_{\tau i}\cdot\boldsymbol{D}_{\tau i}^{-1}\cdot\boldsymbol{C}_{\tau i}, i = 1,2,...,m \tag{9b}$$

Further, we can get the system characteristic equation as following:

$$\det(\lambda\cdot\boldsymbol{I} - \tilde{\boldsymbol{A}}_0 - \sum_{i=1}^{m}\tilde{\boldsymbol{A}}_{\tau i}\cdot e^{-\lambda\cdot\tau_i}) = 0 \tag{10}$$

We denote $\boldsymbol{\Lambda}^{\tau} = (\lambda_1^{\tau}, \lambda_2^{\tau}, ..., \lambda_n^{\tau})$ as the eigenvalue spectrum of the time-delayed system at $(\boldsymbol{x}_0, \boldsymbol{y}_0)$. Similar to the stability criterion without time delay, $(\boldsymbol{x}_0, \boldsymbol{y}_0)$ is stable if all eigenvalues of $\boldsymbol{\Lambda}^{\tau}$ stay in C^-. Otherwise, if we can find some eigenvalue in C^+, $(\boldsymbol{x}_0, \boldsymbol{y}_0)$ is unstable. And, if there exist several eigenvalues on C^0 and all the rest are in C^-, $(\boldsymbol{x}_0, \boldsymbol{y}_0)$ is critical stable.

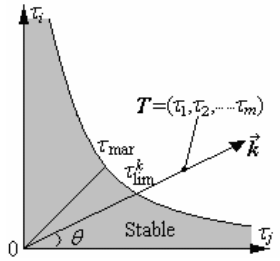


Figure. 1 Illustration of local and global delay margins

2.3. Delay Margin Definition

Vector $\boldsymbol{T} = (\tau_1, \tau_2, ..., \tau_m)$ uniquely yields a direction $\vec{\boldsymbol{k}}$ in the $(\tau_1, \tau_2, ..., \tau_m)$ space as following.

$$\vec{\boldsymbol{k}} = (k_1, k_2, ..., k_m) \tag{11}$$

Where $k_i = \dfrac{\tau_i}{\|\boldsymbol{T}\|}, i = 1,2,3,...,m$ (12)

All points in direction $\vec{\boldsymbol{k}}$ can be expressed as:

$$\boldsymbol{T}^{\boldsymbol{k}} = (k_1, k_2, ..., k_m)\tilde{\tau} = \vec{\boldsymbol{k}}\cdot\tilde{\tau} \tag{13}$$

Especially, $\tilde{\tau}$ equals to $\|\boldsymbol{T}\|$ at point $\boldsymbol{T}$. Provided that when there is no time delay, i.e. $\tilde{\tau} = 0$, power system is stable. Let's increase the value of $\tilde{\tau}$ from zero and calculate the system eigenvalues. If $\tilde{\tau}$ increases to $\tau_{\lim}^k$, we find some eigenvalue on C^0. Before that all eigenvalues stay in C^- and after that there is at least one eigenvalue in C^+, $\tau_{\lim}^k$ is then the local delay margin in direction $\vec{\boldsymbol{k}}$. Furthermore, the global delay margin τ_{mar} is defined as the minimum value in all directions.

$$\tau_{\text{mar}} := \min_{\boldsymbol{k}}(\tau_{\lim}^k) \tag{14}$$

Fig.1 illustrates the local delay margin in direction $\vec{\boldsymbol{k}}$ and the global delay margin.

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III. DELAY MARGIN DETERMINATION METHOD

3.1 Determination of Local Delay Margin in Direction $\vec{k}$

Provided that there is a pair of conjugate eigenvalues on the imaginary axis, say $\lambda_c = \pm iw \in C^0$. Obviously, they satisfy the following equation.

$$iw = \mathrm{eig}(\tilde{\boldsymbol{A}}_0 + \sum_{i=1}^{m} \tilde{\boldsymbol{A}}_{\tau i} \cdot e^{-iw \cdot \tau_i}) \tag{15}$$

Time-delayed constant τ_i in Eq.(15) can be expressed as the format of Eq.(13). Then, substitute it into Eq.(15), we can get:

$$iw = \mathrm{eig}(\tilde{\boldsymbol{A}}_0 + \sum_{i=1}^{m} \tilde{\boldsymbol{A}}_{\tau i} \cdot e^{-ik_i \cdot w \cdot \tilde{\tau}}) = \mathrm{eig}[\Delta(w, \tilde{\tau})]$$

$$\text{Where: } \Delta(w, \tilde{\tau}) = \tilde{\boldsymbol{A}}_0 + \sum_{i=1}^{m} \tilde{\boldsymbol{A}}_{\tau i} \cdot e^{-ik_i \cdot w \cdot \tilde{\tau}} \tag{16}$$

Let $\xi = \tilde{\tau} \cdot w$, Eq.(16) can be expressed as following:

$$\Delta(w, \tilde{\tau}) = \Delta(\xi, \vec{k}) = \tilde{\boldsymbol{A}}_0 + \sum_{i=1}^{m} \tilde{\boldsymbol{A}}_{\tau i} \cdot e^{-ik_i \cdot \xi}$$

Just as we know that $e^{-i\xi}$ changes periodically with ξ, where $\xi \in R$ and the period is 2π. $\Delta(\xi, \vec{k})$ also changes periodically with ξ, and the corresponding period is $\Gamma \cdot 2\pi$. We change ξ in the interval $[0, \Gamma \cdot 2\pi]$ and calculate the eigenvalues of the transformation matrix $\Delta(\xi, \vec{k})$. If there is an eigenvalue, say iw_c , on the imaginary axis at ξ_c , the corresponding critical value of time delay can be gotten as following:

$$\tilde{\tau}_c = \xi_c / \mathrm{abs}(w_c) \tag{17a}$$

There are usually more than one critical time delays in the interval $[0, \Gamma \cdot 2\pi]$.

$$\tilde{\boldsymbol{\tau}}_c = (\tilde{\tau}_{c1}, \tilde{\tau}_{c2}, \tilde{\tau}_{c3}, ...) \tag{17b}$$

The minimum one is the local delay margin in direction $\vec{k}$.

$$\tau_{\lim}^{k} = \min(\tilde{\tau}_{c1}, \tilde{\tau}_{c2}, \tilde{\tau}_{c3}, ...) \tag{17c}$$

And the corresponding vector of time delay is given as below:

$$\boldsymbol{T}_c^k = (\tau_{c1}, \tau_{c2}, ..., \tau_{cm}) = (k_1, k_2, ..., k_m)\tau_{\lim}^{k} \tag{18}$$

Parameter Γ can be determined in this way: Given a direction $\vec{k} = (k_1, k_2, ..., k_m)$, $k_i \in R^+$, if $k_i \in \vec{k}$ is a rational number, it can be expressed as:

$$k_i = M_i / D_i \tag{19}$$

Where M_i , $D_i \in Z^+$ are two prime numbers. If D is the least common multiple of the denominators $(D_1, D_2, ..., D_m)$, then $\Gamma = D$. It is because that the vector of time delay $\boldsymbol{T}$ can be rewritten as following if $k_i \in \vec{k}$ is a rational number.

$$\begin{aligned} \boldsymbol{T} &= (k_1, k_2, ..., k_m)\tilde{\tau} = (M_1/D_1, M_2/D_2, ..., M_m/D_m)\tilde{\tau} \\ &= (M_1', M_2', ..., M_m')\tilde{\tau}' \end{aligned} \tag{20}$$

With $\tilde{\tau}' = \tilde{\tau}/D$, $\xi' = \xi/D$, $M_i' = M_i \cdot D/D_i \in Z^+$, and

$$\Delta(\xi, \vec{k}) = \tilde{\boldsymbol{A}}_0 + \sum_{i=1}^{m} \tilde{\boldsymbol{A}}_{\tau i} \cdot e^{-jM_i' \cdot \xi'} = \Delta(\xi', \vec{k}) \tag{21}$$

It is obvious that $\Delta(\xi', \vec{k})$ changes with ξ' by a period of 2π , and in other words, $\Delta(\xi, \vec{k})$ changes with ξ by a period of $2\pi \cdot D$, i.e. $\Gamma = D$.

Remark 1: When $k_i \in \vec{k}$ is an irrational number, it can be approximated by a rational number under a given round-off error tolerance. e.g. If $k_i = e$, it can be approximated by 2.7182818 with a round-off error less than 10^{-7}.

Remark 2: If $k_j = 0 \in \vec{k}$, i.e. $\vec{k} = (k_1, k_2, ..., k_{j-1}, 0, k_{j+1}, ..., k_m)$, Γ can be determined in the reduced n-1 dimension space $(k_1, k_2, ..., k_{j-1}, k_{j+1}, ..., k_m)$.

3.2 Determination of the Global Delay Margin

In practical application, we are usually interested in the following region.

$$D = (\tau_i \geq 0, i = 1, 2, ..., m) \tag{22}$$

Our aim is to determine the minimum delay margin in region *D*. It is denoted as the global delay margin.

$$\tau_{\mathrm{mar}} = \min_{\boldsymbol{T}(\boldsymbol{k}) \in D} (\tau_{\lim}^{\boldsymbol{k}}) \tag{23}$$

In order to do that, we construct a hyperplane *HP* as following in region *D*:

$$HP = \{\vec{\boldsymbol{p}} | \sum_{i=1}^{m} p_i = 1.0, \vec{\boldsymbol{p}} \in D\} \tag{24}$$

It is clear that each point on the hyperplane will yield a direction in region *D*. And, if all points on the hyperplane are considered, all directions in region *D* are also considered. So, we can use the following algorithm to locate the global delay margin in region *D*.

Step 1: Let $p_i = 0$, and set the incremental step Δp_i , *i*=1, 2,…,*m*.

Step 2: Loop to calculate the delay margin corresponding to the point on hyperplane *HP*.

```
for p1 = 0 : Δp1 : 1.0
  for p2 = 0 : Δp2 : 1.0
    … …
    for pm = 0 : Δpm : 1.0
      if(p ∈ HP) then
        p = (p1, p2, ..., pm);
        k = (p1, p2, ..., pm) / ||p||;
        τ_lim^k = CalcLocalDelayMargin(k);        (25)
      else
        continue;
      end if
    end for
    … …
  end for
end for
```

In the above loop, CalcLocalDelayMargin($\vec{k}$) is called to calculate the local delay margin in direction $\vec{k}$ with the method in Section 3.1.

Step 3: The minimum value of the local delay margin is then the global delay margin, i.e.

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$$\tau_{\mathrm{mar}} = \min_{\vec{p}\in HP}(\tau_{\lim}^{k}) = \min_{\boldsymbol{T}(\vec{\boldsymbol{k}})\in D}(\tau_{\lim}^{\boldsymbol{k}}) \tag{26}$$

IV. CASE STUDIES

Two power systems are employed to validate the given method. One is the single machine and infinite bus (SMIB) system and the other is the WSCC 3-generator-9-bus system.

4.1 SMIB System

- *System Model*

In the SMIB system, synchronous machine is with a field winding and an armature winding. When there is no time delay, system model is given by the following equation. Tab.1 gives parameters used in this paper. More details about the system can be referred to [21].

$$\dot{\delta} = \omega_B \cdot \omega \tag{27a}$$

$$M\dot{\omega} = -D\omega + (P_m - P_G) \tag{27b}$$

$$T'_{d0}\dot{E}' = -E' + (x_d - x'_d)I_d + E_{fd} \tag{27c}$$

$$T_A\dot{E}_{fd} = -K_A(V_G - V_{ref}) - (E_{fd} - E_{fd0}) \tag{27d}$$

$$P_G = E' \cdot V_0 \cdot \sin\delta \cdot B_{ed} \tag{27e}$$

$$V_G = B_{ed} \cdot \sqrt{(x'_d + x_e E'\cos\delta)^2 + (x_e E'\sin\delta)^2} \tag{27f}$$

$$I_d = B_{ed} \cdot (E' - V_0 \cdot \cos\delta) \tag{27g}$$

$$B_{ed} = x_e + x'_d \tag{27h}$$

If the synchronous machine is considered as an equivalent generator controlled by the remote signal in a wide-area environment and there is a time delay in V_G, Eq.(27d) should be replaced by the following time-delayed equation.

$$T_A\dot{E}_{fd} = -K_A(V_G(t-\tau) - V_{ref}) - (E_{fd} - E_{fd0}) \tag{28}$$

- *Delay Margin with D=5.0, K_A=180, P_m=1.0*

Values of the other parameters except D, K_A and P_m are the same as Tab.1. In this condition, equilibrium point of the system is given by Eq.(29). The characteristic equation and the corresponding transformation matrix are given by Eq. (30) and Eq.(31) respectively.

$$\boldsymbol{x}_0 = [0.8045, 0.0000, 1.2492, 1.6196]^T \tag{29a}$$

$$\boldsymbol{y}_0 = [1.000, 1.0521, 0.6\ 175]^T \tag{29b}$$

$$\det(s\boldsymbol{I} - \tilde{\boldsymbol{A}}_0 - \tilde{\boldsymbol{A}}_{\tau 1} \cdot e^{-\tau \cdot s}) = 0 \tag{30}$$

$$\Delta(\xi, \vec{\boldsymbol{k}}) = \tilde{\boldsymbol{A}}_0 + \tilde{\boldsymbol{A}}_{\tau 1} \cdot e^{-j\xi} \tag{31}$$

with,

$$\tilde{\boldsymbol{A}}_0 = \begin{bmatrix} 0 & 376.9911 & 0 & 0 \\ -0.0963 & -0.5000 & -0.0801 & 0 \\ -0.0480 & 0 & -0.1667 & 0.1000 \\ 0 & 0 & 0 & -1.0000 \end{bmatrix} \tag{32}$$

$$\tilde{\boldsymbol{A}}_{\tau 1} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 38.0187 & 0 & -95.2560 & 0 \end{bmatrix} \tag{33}$$

According to Section 3.1, the scanning interval is [0, 2π]. When ξ changes in the interval, we plot the eigenvalue loci of $\Delta(\xi, \vec{\boldsymbol{k}})$ in Fig.2. It can be found that the eigenvalue loci cross the imaginary axis six times. Values of ξ_c, w_c and $\tilde{\tau}_c$ at each crossing point are shown in Tab.2. It is obvious that the delay margin in this condition is 68.249ms.

- *Data Validation*

[13] gave a delay margin determination method based on an optimal procedure. Tab.3-Tab.5 show some comparisons between method of this paper and that of [13] using SMIB system data.

It can be found that the calculation results of this paper and [13] match very well for both the delay margins (τ_{mar}) and imaginary part of the corresponding critical eigenvalue (w_c). Maximum error in three tables is less than 0.01%. Since [13] used an optimal procedure to direct calculate the critical eigenvalue of the time-delayed system so as to determine its delay margin, its results can be used as benchmark. There is almost no error between two results reveals that method of this paper is correct. Comparing with [13], our method is simple and can be applied to the power system with multiple time delays.

Table 1. Parameters of SMIB system used in this paper

M	x_d	x'_d	T'_{d0}	x_e	V_{ref}	V_0	ω_B
10.0	1.0	0.4	10.0	0.5	1.05	1.0	377.0

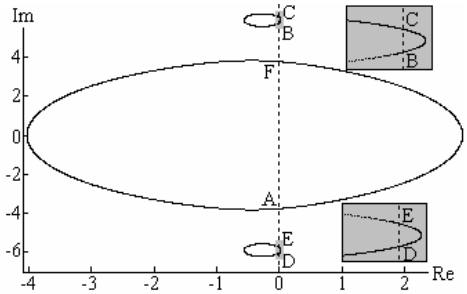


Figure 2. Eigenvalue loci of $\Delta(\xi, \vec{\boldsymbol{k}})$ with D=5.0, K_A=180 and P_m=1.0

Table 2. Point A~point F in figure 2

	ξ_c	w_c	$\tilde{\tau}_c$ (s)
A	0.256982	-3.763327	0.068249
B	0.429770	5.775225	0.074390
C	1.105212	6.015030	0.183726
D	5.178601	-6.015030	0.860855
E	5.854044	-5.775225	1.013565
F	6.026831	3.763327	1.601334

Table 3. Comparison between method of this paper and [13] (D=5.0, K_A=180 and P_m varies between 0-1.0).

P_m		0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
τ_{mar} (ms)	[13]	119.34	118.87	117.48	115.16	111.92	107.73	102.58	96.36	88.91	79.84	68.24
	this paper	119.34	118.88	117.48	115.16	111.92	107.74	102.58	96.36	88.91	79.84	68.25
	error%	0.00%	0.01%	0.00%	0.00%	0.00%	0.01%	0.00%	0.00%	0.00%	0.00%	0.01%
w_c	[13]	3.08187	3.08668	3.10125	3.12594	3.16146	3.20896	3.27028	3.34837	3.44832	3.57960	3.76332
	this paper	3.08187	3.08668	3.10125	3.12594	3.16146	3.20896	3.27028	3.34837	3.44832	3.57960	3.76333
	error%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%

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Table 4. Comparison between method of this paper and [13] (D=5.0, P_m=1.0 and K_A varies between 50~170)

K_A		50	70	90	100	110	120	130	140	150	160	170
τ_{mar} (ms)	[13]	328.50	225.32	169.16	149.59	133.54	120.10	108.63	98.67	89.89	82.02	74.86
	this paper	328.50	225.33	169.16	149.59	133.54	120.10	108.63	98.67	89.89	82.02	74.87
	error%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.01%
w_c	[13]	1.76333	2.14597	2.48045	2.63593	2.78570	2.93098	3.07281	3.21213	3.34985	3.48690	3.62431
	this paper	1.76334	2.14597	2.48045	2.63593	2.78570	2.93098	3.07281	3.21213	3.34985	3.48690	3.62431
	error%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%

Table 5. Comparison between method of this paper and [13] (K_A=180, P_m=1.0 and D varies between 5~190).

D		5	15	30	40	50	60	80	100	120	160	190
τ_{mar} (ms)	[13]	68.24	55.87	44.57	40.81	39.08	38.76	40.74	44.57	49.14	58.51	64.93
	this paper	68.24	55.87	44.57	40.81	39.08	38.76	40.74	44.57	49.14	58.51	64.93
	error%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
w_c	[13]	3.76332	3.71927	3.62362	3.55885	3.49927	3.44609	3.35816	3.29068	3.23869	3.16650	3.13058
	this paper	3.76333	3.71927	3.62362	3.55885	3.49927	3.44609	3.35816	3.29068	3.23870	3.16650	3.13058
	error%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%

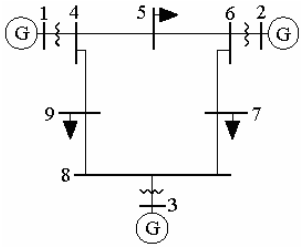


Figure 3. WSCC 3-generator-9-bus system

Table 6. Parameters of the WSCC 3-generator-9-bus system

ω_B	X_{d1}	X'_{d1}	X_{q1}	X'_{q1}	T'_{do1}	T'_{qo1}	H_1
377	0.8958	0.1198	0.8645	0.1969	6.0	0.54	6.4
D_1	T_{A1}	K_{A1}	P_{m1}	V_{ref1}	X_{d2}	X'_{d2}	X_{q2}
0.05	0.02	50.0	1.0	1.03	0.90	0.10	0.85
X'_{q2}	T'_{do2}	T'_{qo2}	H_2	D_2	T_{A2}	K_{A2}	P_{m2}
0.10	8.00	0.25	3.01	0.05	0.02	50.0	1.0
V_{ref2}	P_{L5}	Q_{L5}	P_{L7}	Q_{L7}	P_{L9}	Q_{L9}	
1.03	2.0	1.0	2.0	1.0	2.0	1.0	

Table 7. Eigenvalues of WSCC system without time delay

	Real	Imag
1	-45.8149	
2	-37.5365	
3	-5.0237	
4	-3.2678	
5	-12.6109	±1.3756i
7	-1.3371	±11.5247i
9	-0.0432	±4.9479i

4.2 WSCC 3-Generator-9-Bus System

WSCC system is shown in Fig.3[11]. Generator 3 is treated as an infinite bus. Generator 1 and 2 are considered as two equivalent machines controlled by remote signals. And, there exist time delays in V_{G1} and V_{G2}. Dynamics of two generators are described by the following equations:

$$\dot{\delta}_i = \omega_i - \omega_s \tag{34a}$$

$$2H \cdot \dot{\omega}_i = P_{mi} - (E'_{qi} - X'_{di} \cdot I_{di}) I_{qi} - (E'_{di} + X'_{qi} \cdot I_{qi}) I_{di} - D(\omega_i - \omega_s) \tag{34b}$$

$$T'_{doi} \cdot \dot{E}'_{qi} = -E'_{qi} + (X_{di} - X'_{di}) \cdot I_{di} + E_{fdi} \tag{34c}$$

$$T'_{qoi} \cdot \dot{E}'_{di} = -E'_{di} - (X_{qi} - X'_{qi}) \cdot I_{qi} \tag{34d}$$

$$T_{Ai} \cdot \dot{E}_{fdi} = -E_{fdi} + K_{Ai}(V_{refi} - V_{Gi}(t - \tau_i)) \tag{34e}$$

i=1, 2. Loads are treated as constant power type and modeled as following.

$$P_k = \lambda \cdot P_{k0} = \sum_{j=1}^{n} V_k V_j (G_{kj} \cdot \cos\theta_{kj} + B_{kj} \cdot \sin\theta_{kj}) \tag{35a}$$

$$Q_k = \lambda \cdot Q_{k0} = \sum_{j=1}^{n} V_k V_j (G_{kj} \cdot \sin\theta_{kj} - B_{kj} \cdot \cos\theta_{kj}) \tag{35b}$$

k=1,2,…,9. Tab.6 gives the parameters used in this paper. When time delays are not considered, system eigenvalues are shown in Tab.7. It is clear that system is stable without time delays.

When time delays are considered, matrices in the system characteristic equation are given by Eq.(36). And, local delay margins in various directions are shown in Fig.4. Some typical points in Fig.4 are listed in Tab.8, in which angle θ is defined as $\theta = \tan^{-1}(\tau_2/\tau_1) = \tan^{-1}(k_2/k_1)$.

$$\tilde{A}_0 = \begin{bmatrix} 0 & 377 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.1421 & -0.0039 & -0.0249 & -0.1097 & 0 & 0.1009 & 0 & 0.1202 & 0.0594 & 0 \\ -0.0096 & 0 & -0.2233 & 0.0536 & 0.1667 & 0.1549 & 0 & 0.4965 & 0.0116 & 0 \\ -1.8167 & 0 & 0.2657 & -5.0227 & 0 & 0.9126 & 0 & 0.2903 & 0.7403 & 0 \\ 0 & 0 & -2307.3912 & 962.2607 & -50.0000 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 377 & 0 & 0 & 0 \\ 0.2157 & 0 & 0.2061 & 0.1216 & 0 & -0.3470 & -0.0083 & -0.0708 & -0.2916 & 0 \\ 0.1444 & 0 & 0.3780 & 0.0173 & 0 & -0.0057 & 0 & -0.1092 & 0.0248 & 0.1250 \\ 2.3717 & 0 & 0.4298 & 1.8275 & 0 & -5.5476 & 0 & -0.2416 & -14.2578 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2358.2911 & 829.7367 & -50.0000 \end{bmatrix} \tag{36a}$$

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$$A_{\tau 1}=\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -2578.2817 & 0 & 1618.1539 & -600.5298 & 0 & -191.1358 & 0 & -1016.6381 & 88.5849 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (36b)$$

$$A_{\tau 2}=\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -274.9823 & 0 & -879.6285 & 9.6697 & 0 & -139.8803 & 0 & -43.9157 & -342.6935 & 0 \end{bmatrix} \quad (36c)$$

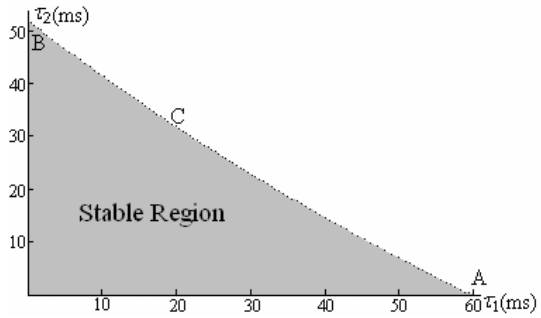


Figure 4. Delay margins of WSCC system in (τ_1, τ_2) space

Table 8. Delay margins in different directions

	θ	τ_2/τ_1	$\tau_1(s)$	$\tau_2(s)$	$\tau_{\lim}^{k}(s)$	w_c
1	0°	0	0.0597	0	0.0597	4.9530
2	10°	0.1763	0.0480	0.0085	0.0487	4.9506
3	20°	0.3640	0.0400	0.0146	0.0426	4.9483
4	30°	0.5774	0.0339	0.0196	0.0391	4.9460
5	40°	0.8391	0.0286	0.0240	0.0374	4.9437
6	45°	1.0000	0.0262	0.0262	0.0370	4.9425
7	50°	1.1918	0.0237	0.0283	0.0370	4.9412
8	60°	1.7321	0.0189	0.0328	0.0379	4.9385
9	70°	2.7475	0.0137	0.0378	0.0402	4.9352
10	80°	5.6712	0.0077	0.0438	0.0445	4.9309
11	90°	∞	0	0.0520	0.0520	4.9248

We can find that system stable region (gray area in Fig.4) is like a triangle. Its right boundary is the curve of local delay margins. We can also find the delay margin that only τ_1 is considered (point A) is larger than the one that only τ_2 is considered (point B). And, the minimum value of the delay margin (i.e. the global delay margin) happens at about θ=48.3° with value τ_{mar}=36.97ms (point C).

V. CONCLUSION

Delay margin is defined as the maximum delayed time that power system can sustain without losing its stability. To determine the delay margin of power system is important for online stability assessment and controller design in wide-area environment. In this paper, a simple approach to determine the delay margin of power system with multiple time delays is given. It traces eigenvalue loci of a transformation matrix in a finite interval to locate all the system critical eigenvalues, which are used to calculate the delay margin. Comparing with previous methods, it is with a lower computation burden and no conservativeness so that it can be used to the stability analysis of bulk power system with time delays. Finally, SMIB and WSCC 3-generator-9-bus system are employed to validate its effectiveness.

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VII. BIOGRAPHIES

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